

Next, Incorporate the Flour: A Recipe for Modeling Noun Incorporation in the LinGO Grammar Matrix

Completed Thesis

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1. Introduction
2. Typology
3. Analysis
4. Implementation
5. Evaluation
6. Conclusion

Introduction

- “Noun Incorporation (NI) is the phenomenon in which a nominal that would otherwise bear a grammatical relation to the verb (such as direct object) is expressed not as an independent noun phrase, but rather as a morphological root that is integrated into the inflected verb to form a kind of composite form.” (Baker et al., 2005)
- IN — an incorporated noun
- Incorporating verb — a verb+IN construction

Typology

- Reduction NI
- Promotion NI
- Modifier NI

- (1) a. ətləg-e qoraŋə-∅ təm-nen
father-erg reindeer-abs kill-3sgA.3sgO
‘The father killed the/a reindeer.’ [ckt] (Polinsky, 1990 via
Gerds, 1998:88)
- b. ətləg-en qaa-nmə-gʔe
father-abs reindeer-kill-3sgS
‘The father killed the reindeer.’ [ckt] (Polinsky, 1990 via
Gerds, 1998:88)

Incorporation triggers intransitive morphosyntactic behavior
(absolutive case marking on the subject, one part agreement on the
verb).

- (2) a. seuan-ide ti-mu-ban
man-sg 1sgA.3sgO-see-past
'I saw the man [tix] (Allen et al., 1984 via Baker et al., 2005:141)
- b. ti-seuan-mu-ban
1sgS/3sgO-man-see-past
'I saw the/a man' [tix] (Allen et al., 1984 via Baker et al., 2005:141)

Incorporation does not trigger intransitive morphosyntactic behavior (same agreement marker used whether the noun is incorporated or not).

- (3) Juan ngilla-waka-fi-y Pedro
Juan buy-cow-3O-ind.3sS Pedro
'Juan bought Pedro's cow.' [arn] (Baker et al., 2005:167)

The verb's direct object is understood to be the possessor of the IN.

- (4) tumg-e ekək kaynə-nmə-nen
friend-erg son-(abs) bear-kill-aor.3sg.3sg
'The friend killed the bear for his son. [ckt] (Polinsky, 1990
via Gerdts, 1998:91)

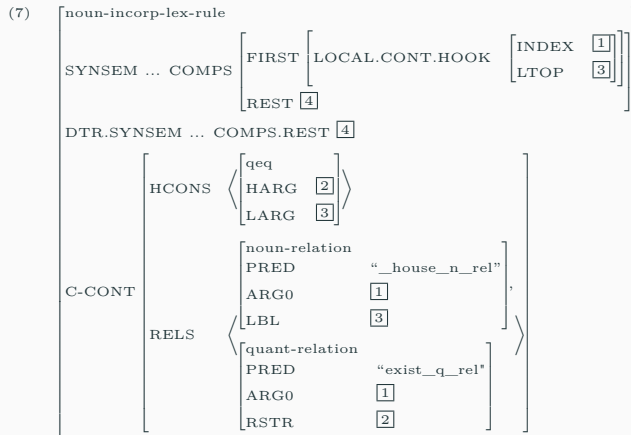
An oblique (ex: benefactive argument) is promoted to the direct object position of the verb when the patient argument is incorporated.

- (5) shateku ni-kuti rabahbót wa-hv-[i]tsy-a-hninu- ki
eight part-ZplS bullhead fact-MsgS-fish-Ø-buy-punc this
rake-niha
my-father
'My father (fish-)bought eight bullheads.' [moh] (Baker,
1996:310 via Baker et al., 2005:149)
- (6) yede a-diru-k'ar-hi
that 2sS/AO-chicken-eat-fut
'You will eat that chicken.' [tix] (Allen et al., 1984:296 via
Baker et al., 2005:150)

Some external material in the direct object position modifies the IN.
This can be a more specific noun (Doubling), or it can be an external
modifier like an adjective or demonstrative (Stranding).

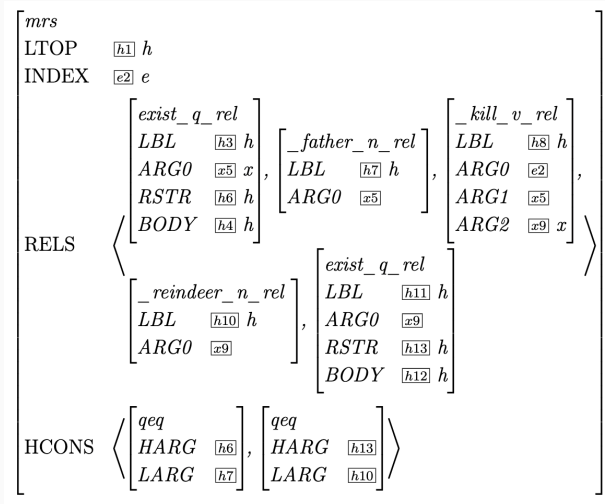
Analysis

- My analysis of NI consists of a pair of lexical rules and in the case of Modifier NI, an additional phrase structure rule.
- First lexical rule shared by all types — does the orthographic and semantic work of incorporating the noun.
- Second of the pair is specific to the type of NI — handles the valency aspect and any other semantic work necessary.
- For Modifier NI instances, the external modifying element goes through a phrase structure rule which primes it to combine with an incorporating verb in the syntax.



ətləg-en qaa-nmə-gʔe
father-abs reindeer-kill-3sgS

‘The father killed the reindeer.’ [ckt] (Polinsky, 1990 via Gerdts, 1998:88)



$$(8) \left[\begin{array}{l} \text{reduce-intrans-lex-rule} \\ \text{SYNSEM ... VAL} \left[\begin{array}{l} \text{SUBJ} \left\langle \begin{array}{l} \text{LOCAL} \left[\begin{array}{l} \text{CONT.HOOK } \boxed{1} \\ \text{CAT} \left[\begin{array}{l} \text{NCORP-MOD } \boxed{2} \\ \text{VAL } \boxed{3} \\ \text{HEAD} \end{array} \right] \end{array} \right] \end{array} \right] \end{array} \right\rangle \\ \text{COMPS } \boxed{4} \end{array} \right] \\ \text{DTR.SYNSEM ...} \left[\begin{array}{l} \text{SUBJ} \left\langle \begin{array}{l} \text{LOCAL} \left[\begin{array}{l} \text{CONT.HOOK } \boxed{1} \\ \text{CAT} \left[\begin{array}{l} \text{NCORP-MOD } \boxed{2} \\ \text{VAL } \boxed{3} \end{array} \right] \end{array} \right] \end{array} \right\rangle \\ \text{COMPS.REST } \boxed{4} \end{array} \right] \end{array} \right]$$

ti-seuan-mu-ban

1sgS/3sgO-man-see-past

‘I saw the/a man’ [tix] (Allen et al., 1984 via Baker et al., 2005:141)

$$\left[\begin{array}{l}
 \text{mrs} \\
 \text{LTOP} \quad \boxed{h1} \ h \\
 \text{INDEX} \quad \boxed{e2} \ e \\
 \\
 \text{RELS} \quad \left\langle \begin{array}{l} \text{[_see_v_rel]} \\ \text{LBL} \quad \boxed{h3} \ h \\ \text{ARG0} \quad \boxed{e2} \\ \text{ARG1} \quad \boxed{x4} \ x \\ \\ \text{ARG2} \quad \boxed{x5} \begin{bmatrix} x \\ \text{SPECI} \quad \text{bool} \\ \text{COG-ST} \quad \text{cog-st} \\ \text{PNG.PER} \quad \text{3rd} \\ \text{PNG.NUM} \quad \text{sing} \end{bmatrix} \end{array} \right\rangle, \begin{array}{l} \text{[_man_n_rel]} \\ \text{LBL} \quad \boxed{h6} \ h \\ \text{ARG0} \quad \boxed{x5} \end{array}, \begin{array}{l} \text{[exist_q_rel]} \\ \text{LBL} \quad \boxed{h7} \ h \\ \text{ARG0} \quad \boxed{x5} \\ \text{RSTR} \quad \boxed{h9} \ h \\ \text{BODY} \quad \boxed{h8} \ h \end{array} \right\rangle \\
 \\
 \text{HCONS} \quad \left\langle \begin{array}{l} \text{[geq]} \\ \text{HARG} \quad \boxed{h9} \\ \text{LARG} \quad \boxed{h6} \end{array} \right\rangle \oplus \text{null} \\
 \\
 \text{ICONS} \quad \left\langle \begin{array}{l} \text{[non-focus]} \\ \text{LARG} \quad \boxed{e2} \\ \text{RARG} \quad \boxed{x5} \end{array}, \begin{array}{l} \text{[non-focus]} \\ \text{LARG} \quad \boxed{e2} \\ \text{RARG} \quad \boxed{x4} \end{array} \right\rangle \oplus \text{null}
 \end{array} \right]$$

$$(9) \left[\begin{array}{l} \text{reduce-trans-lex-rule} \\ \text{SYNSEM.LOCAL.CAT.VAL} \\ \text{DTR.SYNSEM.LOCAT.CAT.VAL} \end{array} \left[\begin{array}{l} \text{SUBJ} \quad \boxed{1} \\ \text{COMPS} \quad \boxed{2} \langle [\text{OPT} \quad +], \dots \rangle \\ \text{SUBJ} \quad \boxed{1} \\ \text{COMPS} \quad \boxed{2} \end{array} \right] \right]$$

Juan ngilla-waka-fi-y Pedro

Juan buy-cow-3O-ind.3sS Pedro

‘Juan bought Pedro’s cow.’ [arn] (Baker et al., 2005:167)

<i>mrs</i>	
LTOP	h1 <i>h</i>
INDEX	e2 <i>e</i>
RELS	$\left\langle \begin{array}{l} \text{exist_q_rel} \\ \text{LBL} \quad \text{h3} \quad h \\ \text{ARG0} \quad \text{x5} \quad x \\ \text{RSTR} \quad \text{h6} \quad h \\ \text{BODY} \quad \text{h4} \quad h \end{array} , \begin{array}{l} \text{_juan_n_rel} \\ \text{LBL} \quad \text{h7} \quad h \\ \text{ARG0} \quad \text{x5} \end{array} , \begin{array}{l} \text{_buy_v_rel} \\ \text{LBL} \quad \text{h8} \quad h \\ \text{ARG0} \quad \text{e2} \\ \text{ARG1} \quad \text{x5} \\ \text{ARG2} \quad \text{x9} \quad x \end{array} \right\rangle$
	$\left\langle \begin{array}{l} \text{exist_q_rel} \\ \text{LBL} \quad \text{h10} \quad h \\ \text{ARG0} \quad \text{x9} \\ \text{RSTR} \quad \text{h12} \quad h \\ \text{BODY} \quad \text{h11} \quad h \end{array} , \begin{array}{l} \text{poss_rel} \\ \text{LBL} \quad \text{h13} \quad h \\ \text{ARG0} \quad \text{e14} \quad e \\ \text{ARG1} \quad \text{x9} \\ \text{ARG2} \quad \text{x15} \quad x \end{array} , \begin{array}{l} \text{exist_q_rel} \\ \text{LBL} \quad \text{h16} \quad h \\ \text{ARG0} \quad \text{x15} \\ \text{RSTR} \quad \text{h18} \quad h \\ \text{BODY} \quad \text{h17} \quad h \end{array} \right\rangle$
	$\begin{array}{l} \text{_pedro_n_rel} \\ \text{LBL} \quad \text{h19} \quad h \\ \text{ARG0} \quad \text{x15} \end{array} , \begin{array}{l} \text{_cow_n_rel} \\ \text{LBL} \quad \text{h13} \\ \text{ARG0} \quad \text{x9} \end{array}$
	$\left\langle \begin{array}{l} \text{qeq} \\ \text{HARG} \quad \text{h6} \\ \text{LARG} \quad \text{h7} \end{array} , \begin{array}{l} \text{qeq} \\ \text{HARG} \quad \text{h12} \\ \text{LARG} \quad \text{h13} \end{array} , \begin{array}{l} \text{qeq} \\ \text{HARG} \quad \text{h18} \\ \text{LARG} \quad \text{h19} \end{array} \right\rangle$
HCONS	

(10)
$$\left[\begin{array}{l} \text{promote-poss-lex-rule} \\ \\ \text{SYNSEM ... VAL} \left[\begin{array}{l} \text{SUBJ } \boxed{1} \\ \text{COMPS} \left\langle \begin{array}{l} \text{OPT } - \\ \text{LOCAL} \left[\begin{array}{l} \text{CAT} \left[\begin{array}{l} \text{NCORP-MOD } \boxed{2} \\ \text{VAL.SPR } () \\ \text{HEAD noun} \\ \text{CONT.HOOK.INDEX } \boxed{3} \end{array} \right] \end{array} \right] \end{array} \right\rangle, \boxed{6} \end{array} \right] \\ \\ \text{DTR.SYNSEM ... VAL} \left[\begin{array}{l} \text{SUBJ } \boxed{1} \\ \text{COMPS} \left\langle \text{LOCAL} \left[\begin{array}{l} \text{CAT.NCORP-MOD } \boxed{2} \\ \text{CONT.HOOK} \left[\begin{array}{l} \text{INDEX } \boxed{4} \\ \text{LTOP } \boxed{5} \end{array} \right] \end{array} \right] \end{array} \right\rangle, \boxed{6} \end{array} \right] \\ \\ \text{C-CONT.RELS} \left\langle \begin{array}{l} \text{arg12-ev-relation} \\ \text{PRED "poss_rel"} \\ \text{LBL } \boxed{5} \\ \text{ARG1 } \boxed{4} \\ \text{ARG2 } \boxed{3} \end{array} \right\rangle \end{array} \right]$$

tumg-e ekək kaynə-nmə-nen
friend-erg son-(abs) bear-kill-aor.3sg.3sg

‘The friend killed the bear for his son. [ckt] (Polinsky, 1990 via Gerdts, 1998:91)

<i>mrs</i>	
LTOP	h1 <i>h</i>
INDEX	e2 <i>e</i>
RELS	$\left\langle \begin{array}{l} \textit{exist_q_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h3}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{x5}} \textit{x} \\ \textit{RSTR} \quad \textstyle \boxed{\text{h6}} \textit{h} \\ \textit{BODY} \quad \textstyle \boxed{\text{h4}} \textit{h} \end{array}, \begin{array}{l} \textit{_friend_n_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h7}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{x5}} \end{array}, \begin{array}{l} \textit{exist_q_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h8}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{x10}} \textit{x} \\ \textit{RSTR} \quad \textstyle \boxed{\text{h11}} \textit{h} \\ \textit{BODY} \quad \textstyle \boxed{\text{h9}} \textit{h} \end{array} \right\rangle$
	$\left\langle \begin{array}{l} \textit{_son_n_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h12}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{x10}} \end{array}, \begin{array}{l} \textit{_kill_v_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h13}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{e2}} \\ \textit{ARG1} \quad \textstyle \boxed{\text{x5}} \\ \textit{ARG2} \quad \textstyle \boxed{\text{x14}} \textit{x} \end{array}, \begin{array}{l} \textit{_bear_n_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h15}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{x14}} \end{array} \right\rangle$
	$\left\langle \begin{array}{l} \textit{exist_q_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h16}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{x14}} \\ \textit{RSTR} \quad \textstyle \boxed{\text{h18}} \textit{h} \\ \textit{BODY} \quad \textstyle \boxed{\text{h17}} \textit{h} \end{array}, \begin{array}{l} \textit{benefactive_rel} \\ \textit{LBL} \quad \textstyle \boxed{\text{h19}} \textit{h} \\ \textit{ARG0} \quad \textstyle \boxed{\text{e20}} \textit{e} \\ \textit{ARG1} \quad \textstyle \boxed{\text{e2}} \\ \textit{ARG2} \quad \textstyle \boxed{\text{x10}} \end{array} \right\rangle$
HCONS	$\left\langle \begin{array}{l} \textit{qeq} \\ \textit{HARG} \quad \textstyle \boxed{\text{h6}} \\ \textit{LARG} \quad \textstyle \boxed{\text{h7}} \end{array}, \begin{array}{l} \textit{qeq} \\ \textit{HARG} \quad \textstyle \boxed{\text{h11}} \\ \textit{LARG} \quad \textstyle \boxed{\text{h12}} \end{array}, \begin{array}{l} \textit{qeq} \\ \textit{HARG} \quad \textstyle \boxed{\text{h18}} \\ \textit{LARG} \quad \textstyle \boxed{\text{h15}} \end{array} \right\rangle$

$$\begin{array}{l}
 (11) \quad \left[\begin{array}{l}
 \text{promote-obl-lex-rule} \\
 \\
 \text{SYNSEM} \left[\begin{array}{l}
 \text{CONT.HOOK.LTOP} \boxed{7} \\
 \dots \text{VAL} \left[\begin{array}{l}
 \text{SUBJ} \boxed{1} \\
 \text{COMPS} \left\langle \begin{array}{l}
 \text{OPT } - \\
 \text{LOCAL} \left[\begin{array}{l}
 \text{CAT} \left[\begin{array}{l}
 \text{NCORP-MOD} \boxed{6} \\
 \text{VAL.SPR} \langle \rangle \\
 \text{HEAD} \boxed{4} \text{ noun}
 \end{array} \right] \\
 \text{CONT.HOOK.INDEX} \boxed{3}
 \end{array} \right]
 \end{array} \right]
 \end{array} \right]
 \end{array} \right]
 \end{array} \right]
 \end{array}
 \right. \\
 \\
 \text{DTR } \dots \text{ LOCAL} \left[\begin{array}{l}
 \text{CAT.VAL} \left[\begin{array}{l}
 \text{SUBJ} \boxed{1} \\
 \text{COMPS} \left\langle \text{LOCAL.CAT} \left[\begin{array}{l}
 \text{NCORP-MOD} \boxed{6} \\
 \text{HEAD} \boxed{4}
 \end{array} \right], \boxed{5} \right\rangle
 \end{array} \right] \\
 \text{CONT.HOOK.INDEX} \boxed{2}
 \end{array} \right]
 \end{array}
 \right. \\
 \\
 \text{C-CONT.RELS} \left\langle \begin{array}{l}
 \text{arg12-ev-relation} \\
 \text{LBL} \quad \boxed{7} \\
 \text{ARG1} \quad \boxed{2} \\
 \text{ARG2} \quad \boxed{3}
 \end{array} \right\rangle
 \end{array} \right]
 \end{array}
 \right.
 \end{array}$$

shateku ni-kuti rabahbót wa-hv-[i]tsy-a-hninu- ki rake-niha
eight part-ZplS bullhead fact-MsgS-fish-Ø-buy-punc this my-father
'My father (fish-)bought eight bullheads.' [moh] (Baker, 1996:310 via
Baker et al., 2005:149)

<i>mrs</i>			
LTOP h1 h			
INDEX e2 e			
	$\begin{bmatrix} \text{type_rel} \\ \text{LBL} \quad \text{h3} \quad h \\ \text{ARG0} \quad \text{e4} \quad e, \\ \text{ARG1} \quad \text{x6} \quad x \\ \text{ARG2} \quad \text{x5} \quad x \end{bmatrix}$	$\begin{bmatrix} \text{exist_q_rel} \\ \text{LBL} \quad \text{h7} \quad h \\ \text{ARG0} \quad \text{x5} \\ \text{RSTR} \quad \text{h9} \quad h \\ \text{BODY} \quad \text{h8} \quad h \end{bmatrix}$	$\begin{bmatrix} \text{_eight_a_rel} \\ \text{LBL} \quad \text{h10} \quad h \\ \text{ARG0} \quad \text{e11} \quad e \\ \text{ARG1} \quad \text{x5} \end{bmatrix}$
RELS	$\left\langle \begin{bmatrix} \text{_bullhead_n_rel} \\ \text{LBL} \quad \text{h10} \\ \text{ARG0} \quad \text{x5} \end{bmatrix}, \begin{bmatrix} \text{_buy_v_rel} \\ \text{LBL} \quad \text{h12} \quad h \\ \text{ARG0} \quad \text{e2} \\ \text{ARG1} \quad \text{x13} \quad x \\ \text{ARG2} \quad \text{x6} \end{bmatrix}, \begin{bmatrix} \text{exist_q_rel} \\ \text{LBL} \quad \text{h14} \quad h \\ \text{ARG0} \quad \text{x6} \\ \text{RSTR} \quad \text{h16} \quad h \\ \text{BODY} \quad \text{h15} \quad h \end{bmatrix} \right\rangle$		
	$\begin{bmatrix} \text{_fish_n_rel} \\ \text{LBL} \quad \text{h3} \\ \text{ARG0} \quad \text{x6} \end{bmatrix}$	$\begin{bmatrix} \text{_father_n_rel} \\ \text{LBL} \quad \text{h17} \quad h \\ \text{ARG0} \quad \text{x13} \end{bmatrix}$	$\begin{bmatrix} \text{pron_rel} \\ \text{LBL} \quad \text{h18} \quad h \\ \text{ARG0} \quad \text{x19} \quad x \end{bmatrix}$
	$\begin{bmatrix} \text{exist_q_rel} \\ \text{LBL} \quad \text{h20} \quad h \\ \text{ARG0} \quad \text{x13} \\ \text{RSTR} \quad \text{h22} \quad h \\ \text{BODY} \quad \text{h21} \quad h \end{bmatrix}$	$\begin{bmatrix} \text{poss_rel} \\ \text{LBL} \quad \text{h17} \\ \text{ARG0} \quad \text{e23} \quad e, \\ \text{ARG1} \quad \text{x13} \\ \text{ARG2} \quad \text{x19} \end{bmatrix}$	$\begin{bmatrix} \text{exist_q_rel} \\ \text{LBL} \quad \text{h24} \quad h \\ \text{ARG0} \quad \text{x19} \\ \text{RSTR} \quad \text{h26} \quad h \\ \text{BODY} \quad \text{h25} \quad h \end{bmatrix}$
HCONS	$\left\langle \begin{bmatrix} \text{qeq} \\ \text{HARG} \quad \text{h9} \\ \text{LARG} \quad \text{h10} \end{bmatrix}, \begin{bmatrix} \text{qeq} \\ \text{HARG} \quad \text{h16} \\ \text{LARG} \quad \text{h3} \end{bmatrix}, \begin{bmatrix} \text{qeq} \\ \text{HARG} \quad \text{h22} \\ \text{LARG} \quad \text{h17} \end{bmatrix}, \begin{bmatrix} \text{qeq} \\ \text{HARG} \quad \text{h26} \\ \text{LARG} \quad \text{h18} \end{bmatrix} \right\rangle$		

(12)

$$\left[\begin{array}{l} \text{double-lex-rule} \\ \\ \text{SYNSEM ... VAL} \left[\begin{array}{l} \text{SUBJ} \quad \boxed{1} \\ \text{COMPS} \left\langle \begin{array}{l} \text{OPT -} \\ \text{LOCAL} \left[\begin{array}{l} \text{CAT} \left[\begin{array}{l} \text{NCORP-MOD +} \\ \text{HEAD noun} \end{array} \right] \\ \text{CONT.HOOK} \left[\begin{array}{l} \text{LTOP} \quad \boxed{2} \\ \text{INDEX} \quad \boxed{3} \end{array} \right] \end{array} \right] \end{array} \right] \end{array} \right] \end{array} \right] \\ \\ \text{DTR ... VAL} \left[\begin{array}{l} \text{SUBJ} \quad \boxed{1} \\ \text{COMPS} \left\langle \begin{array}{l} \text{LOCAL.CONT.HOOK} \left[\begin{array}{l} \text{LTOP} \quad \boxed{2} \\ \text{INDEX} \quad \boxed{3} \end{array} \right] \end{array} \right] \end{array} \right] \end{array} \right] \\ \\ \text{C-CONT} \left[\begin{array}{l} \text{RELS} \quad \langle \rangle \\ \text{HCONS} \quad \langle \rangle \\ \text{ICONS} \quad \langle \rangle \end{array} \right] \end{array} \right]$$

(13)

$$\left[\begin{array}{l} \text{type-ni-mod-phrase} \\ \\ \text{SYNSEM.LOCAL} \\ \\ \text{HEAD-DTR ... LOCAL} \\ \\ \text{C-CONT} \end{array} \left[\begin{array}{l} \text{CAT} \left[\begin{array}{l} \text{VAL} \left[\begin{array}{l} \text{NCORP-MOD} \quad + \end{array} \right] \end{array} \right] \left[\begin{array}{l} \text{CONT.HOOK.INDEX} \quad \boxed{2} \end{array} \right] \\ \\ \text{CAT} \left[\begin{array}{l} \text{HEAD} \quad \text{noun} \\ \text{VAL} \quad \boxed{1} \left[\begin{array}{l} \text{SPR} \quad \langle \rangle \end{array} \right] \\ \text{NCORP-MOD} \quad - \end{array} \right] \left[\begin{array}{l} \text{CONT.HOOK.INDEX} \quad \boxed{2} \end{array} \right] \\ \\ \text{RELS} \left\langle \begin{array}{l} \text{arg12-ev-relation} \\ \text{PRED} \quad \text{"type_rel"} \\ \text{LBL} \quad \boxed{3} \\ \text{ARG1} \quad \boxed{4} \\ \text{ARG2} \quad \boxed{2} \end{array} \right\rangle \\ \\ \text{ICONS} \quad \langle \rangle \\ \text{HCONS} \quad \langle \rangle \\ \\ \text{HOOK} \left[\begin{array}{l} \text{LTOP} \quad \boxed{3} \\ \text{INDEX} \quad \boxed{4} \end{array} \right] \end{array} \right] \right]$$

yede a-diru-k'ar-hi
that 2sS/AO-chicken-eat-fut

‘You will eat that chicken.’ [tix] (Allen et al., 1984:296 via Baker et al., 2005:150)

mrs

LTOP $\boxed{h1} \ h$

INDEX $\boxed{e2} \ e$

RELS $\left\langle \begin{bmatrix} \textit{exist_q_rel} \\ \textit{LBL} \quad \boxed{h3} \ h \\ \textit{ARG0} \quad \boxed{x5} \ x \\ \textit{RSTR} \quad \boxed{h6} \ h \\ \textit{BODY} \quad \boxed{h4} \ h \end{bmatrix}, \begin{bmatrix} \textit{_that_a_rel} \\ \textit{LBL} \quad \boxed{h7} \ h \\ \textit{ARG0} \quad \boxed{e8} \ e \\ \textit{ARG1} \quad \boxed{x5} \end{bmatrix}, \begin{bmatrix} \textit{_eat_v_rel} \\ \textit{LBL} \quad \boxed{h9} \ h \\ \textit{ARG0} \quad \boxed{e2} \\ \textit{ARG1} \quad \boxed{x10} \ x \\ \textit{ARG2} \quad \boxed{x5} \end{bmatrix} \right\rangle$

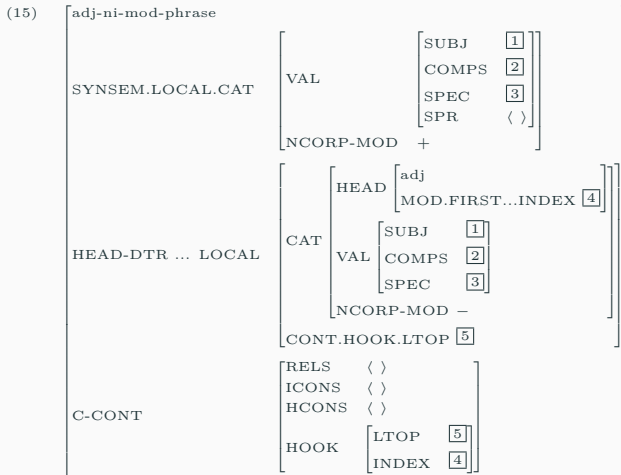
$\begin{bmatrix} \textit{_chicken_n_rel} \\ \textit{LBL} \quad \boxed{h7} \\ \textit{ARG0} \quad \boxed{x5} \end{bmatrix}$

HCONS $\left\langle \begin{bmatrix} \textit{qeq} \\ \textit{HARG} \quad \boxed{h6} \\ \textit{LARG} \quad \boxed{h7} \end{bmatrix} \right\rangle$

ICONS $\left\langle \begin{bmatrix} \textit{non-focus} \\ \textit{LARG} \quad \boxed{e2} \\ \textit{RARG} \quad \boxed{x10} \end{bmatrix} \right\rangle$

(14)

$$\left[\begin{array}{l} \text{strand-lex-rule} \\ \\ \text{SYNSEM ... VAL} \\ \\ \text{DTR ... VAL} \\ \\ \text{C-CONT} \end{array} \left[\begin{array}{l} \left[\begin{array}{l} \text{SUBJ} \left\langle \begin{array}{l} \text{LOCAL} \left[\begin{array}{l} \text{CONT.HOOK } \boxed{1} \\ \text{CAT} \left[\begin{array}{l} \text{NCORP-MOD } \boxed{2} \\ \text{VAL } \boxed{3} \\ \text{HEAD +np} \end{array} \right] \end{array} \right] \right\rangle \end{array} \right. \\ \left. \text{COMPS} \left\langle \begin{array}{l} \text{OPT -} \\ \text{LOCAL} \left[\begin{array}{l} \text{CAT} \left[\begin{array}{l} \text{NCORP-MOD +} \\ \text{HEAD +jp} \end{array} \right] \\ \text{CONT.HOOK} \left[\begin{array}{l} \text{LTOP } \boxed{5} \\ \text{INDEX } \boxed{6} \end{array} \right] \end{array} \right] \end{array} \right\rangle, \boxed{4} \right\rangle \end{array} \right] \\ \left[\begin{array}{l} \text{SUBJ} \left\langle \begin{array}{l} \text{LOCAL} \left[\begin{array}{l} \text{CONT.HOOK } \boxed{1} \\ \text{CAT} \left[\begin{array}{l} \text{NCORP-MOD } \boxed{2} \\ \text{VAL } \boxed{3} \end{array} \right] \end{array} \right] \right\rangle \\ \text{COMPS} \left\langle \begin{array}{l} \text{LOCAL.CONT.HOOK} \left[\begin{array}{l} \text{LTOP } \boxed{5} \\ \text{INDEX } \boxed{6} \end{array} \right] \end{array} \right\rangle, \boxed{4} \right\rangle \end{array} \right] \\ \left[\begin{array}{l} \text{RELS} \quad \langle \rangle \\ \text{HCONS} \quad \langle \rangle \\ \text{ICONS} \quad \langle \rangle \end{array} \right] \end{array} \right] \end{array} \right]$$



Implementation

- My analysis consists of both the common NI lexical rule as well as a type-specific lexical rule.
- I wanted to make the user experience straightforward.
- Instead of having the user define two position classes on the morphology subpage, I have them define one which corresponds to the general NI lexical rule.
- User choices on a separate NI subpage give me the information needed to create the second lexical rule behind the scenes.

Noun Incorporation ([documentation](#))

Noun Incorporation is the process of morphologically incorporating a noun, usually the direct object of the verb, into the verb itself resulting in a complex verb.

☐ Noun Incorporation occurs in this language.

☒ The direct object of the verb incorporates and there is no other overt expression of this argument.

If so, how does the verb behave?

- ☐ The verb appears transitive (i.e. transitive morphology, case)
☒ The verb appears intransitive (i.e. intransitive morphology, case)

If intransitive, does the case marked on the subject change?

☐ No

☒ Yes. New case:

▼

☐ ergative

☒ absolutive

Are there any position classes of NI should not co-occur with?

Add a Forbid constraint

☐ The direct object of the verb incorporates and another argument (such as a possessor, locative, instrumental, etc.) becomes the new direct object.

If so, what type of noun is promoted to object position?

☐ The possessor of the incorporated noun is raised to object position

☐ An oblique argument is promoted to object position. You may define a predicate name for this process. Predicate:

Are there any position classes that this type of NI should not co-occur with?

[Add a Forbid constraint](#)

☐ Elements of the incorporated noun appear outside of the verb. For example, a co-referent noun that is more specific than the incorporated noun, modifiers or demonstratives of the incorporated noun, or a copy of the incorporated noun itself.

If so, what elements can be stranded?

☐ A more specific noun or doubled noun

Are there any position classes that this type of NI should not co-occur with?

Add a Forbid constraint

☐ A modifier (ex: adjective, demonstrative, quantifier)

Are there any position classes that this type of NI should not co-occur with?

Add a Forbid constraint

Verb Inflection

▼ noun_incorporation (verb-pc1)



Verb Position Class 1:

Position Class Name: noun_incorporation

Obligatorily occurs: ☐

Appears as a prefix or suffix: Prefix ▼

Possible inputs: ▼

Morphotactic Constraints:

Add a Require constraint

Add a Forbid constraint

Lexical Rule Types that appear in this Position Class:

► verb-pc1_lrt1

Add a Lexical Rule Type

Add an Incorporated Stem Lexical Rule Type

▼ incorporated_noun (verb-pc1_is-lrt1)

☒ Incorporated Stem Lexical Rule Type 1:

Name:

Supertypes: ▼

Features:

☒ Name: Value: ▼ Specified on: ▼

Morphotactic Constraints:

Lexical Rule Instances:

☒ Instance 1

☐ No affix

☒ Affix spelled

Predicate:

Evaluation

Illustrative Languages

Language	Reduction NI			Promotion NI		Modifier NI	
	trans	intrans same	intrans different	promote poss	promote obl	strand	double
Chukchi			✓	✓	✓		
S. Tiwa	✓					✓	
Mapudungun		✓		✓			
Mohawk		✓			✓	✓	✓
Tongan			✓			✓	

Illustrative language choices on NI subpage

Language	WO	case system	subj/obj agreement	transitivity marking	valchange ops
Chukchi	free	erg/abs, locative	subj+obj	no	applicative
S. Tiwa	free	(nom/acc), locative	subj+obj	no	passive
Mapudungun	SVO	none	subj, obj	no	applicative
Mohawk	free	none	subj+obj	no	
Tongan	free	erg/abs	none	no	

Illustrative language additional choices for testing library interaction

Language	Coverage	Overgeneration	Ambiguity
Chukchi	100%	0%	1.43
S. Tiwa	92.3%	0%	1.58
Mapudungun	100%	25%	1
Mohawk	100%	50%	1.4
Tongan	100%	0%	1

Illustrative language grammar performance

Pseudo-languages

Language	Reduction NI			Promotion NI		Modifier NI	
	trans	intrans same	intrans different	promote poss	promote obl	strand	double
PL1	✓			✓	✓	✓	✓
PL2		✓		✓	✓	✓	✓
PL3			✓	✓	✓	✓	✓
PL4	✓			✓	✓	✓	✓
PL5			✓	✓	✓	✓	✓
PL6	✓			✓	✓	✓	✓

Pseudo-language choices on NI subpage

Pseudo-languages

Language	WO	case system	subj/obj agreement	transitivity marking	valchange ops
PL1	SOV	erg/abs	subj, obj	yes	
PL2	SVO	nom/acc	subj, obj	yes	
PL3	VSO	erg/abs	subj, obj	yes	
PL4	OSV	tripartite	subj+obj	yes	
PL5	OVS	tripartite	subj, obj	yes	
PL6	VOS	(nom/acc), benefactive, instrumental, causative	subj+obj	no	passive, applicative, causative

Pseudo-language additional choices for testing library interaction

Language	Coverage	Overgeneration	Ambiguity
PL1	100%	5%	1.5
PL2	100%	0%	1.57
PL3	100%	0%	1.57
PL4	100%	5%	1.43
PL5	100%	0%	1.57
PL6	100%	0%	2.5

Pseudo-language grammar performance

Held-out Languages

Language	Reduction NI			Promotion NI		Modifier NI	
	trans	intrans same	intrans different	promote poss	promote obl	strand	double
Apurinã	✓			✓		✓	✓
Bribri			✓	✓	✓		
Inuktitut			✓		✓	✓	✓
Moloko		✓					
Yaqui		✓					

Held-out language choices on NI subpage

Held-out Languages

Language	WO	case system	subj/obj agreement	transitivity	valchange ops
Apurinã	VSO	(nom/acc), goal	subj, obj	no	
Bribri	free	erg/abs, intrs, com, loc, ben	none	imperfective	obj-add
Inuktitut	SOV	erg/abs, mod, allat	subj+obj	no	passive, antipassive
Moloko	SVO	(nom/acc), loc, dat	subj, ind obj	no	obj-add
Yaqui	SOV	nom/acc	none	yes	passive

Held-out language additional choices for testing library interaction

Language	Coverage	Overgeneration	Ambiguity
Apurinã	100%	23.1%	2.1
Bribri	100%	7.1%	2.4
Inuktitut	77.8%	33.3%	1.62
Moloko	100%	0%	1.2
Yaqui	100%	12.5%	1

Held-out language grammar performance

Conclusion

Since last DELPH-IN...

- Updated parts of my analysis during development, including adding a new type of Promotion NI for promoted oblique arguments
- Finished the implementation on the backend and frontend, mapping user answers from the customization page to customized grammar fragments
- Created six psuedo-languages to (more-fully) test each combination of choices
- Found five unseen languages to test my system after freezing development

- Successful interactions with other Matrix libraries
 - For example, interaction of NI and passivization from the Valence Change library (Curtis, 2018).
 - Incorporation of a demoted subject in Southern Tiwa:

- (16) a. khwienide edeureban kanideba
khwien-ide edeuri-e-ban kan-ide-ba
dog-sg kick-pass-past horse-sg-inst
'The dog was kicked by the horse. [tix] (Allen et al., 1984:302)
- b. khwienide kanedeurieban
khwien-ide kan-edeuri-e-ban
dog-sg horse-kick-pass-past
'The dog was kicked by the horse. [tix] (Allen et al., 1984:302)

Future Work

- Incorporate an analysis of information structure as related to NI
- Expand to other types of NI (subject incorporation, morphologically complex incorporation, oblique argument incorporation)
- Improve customizability:
 - one failed interaction with the valence change library would have been avoided had I allowed users to state whether promoted or doubled arguments could be dropped
 - allow for users to specify POS for promoted and doubled elements (I constrained these values in my rules)
 - allow for the potential of both the transitive and intransitive types of Reduction NI to be applicable for a given language
- Better constrain the NCORP-MOD feature — ambiguity due to its value being underspecified in unexpected places
- Build out functionality to ensure that the doubled noun actually is a type of the IN (i.e./ to rule out sentences like ‘I fish-caught grapes.’)

Thank you!

References

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